

Coding & Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub Repository – Emotion Detection & Learning Support Engine
Programming Language(s)	Python
Framework(s) Used	Streamlit, TensorFlow, Hugging Face Transformers
Key Features Implemented	Text preprocessing, BiLSTM emotion detection, BERT emotion classification, mixed emotion detection, AI-powered learning guidance using Google Gemini API, emotion analytics dashboard, session history management, CSV export, interactive Streamlit interface.
Pending / Incomplete Features	User authentication, database integration (MySQL/MongoDB), cloud deployment, real-time emotion monitoring, mobile application support.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Create a virtual environment (python -m venv .venv). 3. Activate the virtual environment. 4. Install dependencies using pip install -r requirements.txt . 5. Ensure trained models are placed in the models/ directory. 6. Configure the Gemini API key in the .env file. 7. Run the application using streamlit run app.py .

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes.	Yes
2	Meaningful variable and function names are used.	Yes
3	Code includes comments/documentation where necessary.	Yes
4	Error handling is implemented for critical operations.	Yes
5	The application runs without critical errors.	Yes
6	Code is committed to a version control repository.	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate **data**, **models**, **notebooks**, **src**, and **Streamlit application** layers. The application integrates **BiLSTM**, **BERT**, and **Google Gemini AI** to provide emotion detection and personalized learning guidance. The solution has been tested for preprocessing, model prediction, and AI response generation and is ready for demonstration, with future enhancements planned for cloud deployment and user authentication.